

**A Project Report
on**

IOT - BASED PREDICTIVE IRRIGATION SCHEDULING USING VAPOUR PRESSURE DEFICIT(VPD) AND SOLAR FLUX FUSION

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In partial fulfillment of the requirements for the award of the degree of

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**DEPARTMENT OF
ELECTRONICS AND COMMUNICATION ENGINEERING**

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2025-2026**

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BONAFIDE CERTIFICATE

This is to certify that the project entitled “ **IoT-BASED PREDICTIVE IRRIGATION SCHEDULING USING VAPOUR PRESSURE DEFICIT (VPD) AND SOLAR FLUX FUSION** ” is being submitted by **P.HARI CHANDANA (22711A0446), S.SRI LAKSHMI (22711A0454), MD.ERAM ZULEKHA (22711A0435), SK.AFREEN (23715A0404)** towards partial fulfillment of the requirements for the award of degree of Bachelor of Technology in the Department of **ELECTRONICS AND COMMUNICATION ENGINEERING**, NARAYANA ENGINEERING COLLEGE (A):: NELLORE, an Autonomous college under JNTUA Ananthapuramu is bonafide work carried out under my guidance and supervision.

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DECLARATION

We hereby declare that the project work entitled “ **IoT-Based Predictive Irrigation Scheduling Using Vapour Pressure Deficit (VPD) And Solar Flux Fusion**” has been done by us under the guidance of **Mr. V. Praveen Kumar, M.Tech ,(Ph.D) Assistant Professor**, Department of ECE , this project work has been submitted to the **Narayana Engineering College, (Autonomous), Nellore** as partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology**.

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ABSTRACT

This project presents an IoT-based predictive irrigation system that utilizes Vapor Pressure Deficit (VPD) and solar flux fusion to improve water management in agriculture. Traditional irrigation systems rely on fixed soil moisture thresholds, which often lead to inefficient water usage due to their inability to adapt to dynamic environmental conditions. In contrast, the proposed system integrates real-time atmospheric and soil data using advanced sensors, including a BME280 atmospheric sensor and a BH1750 light intensity sensor, connected to an ESP32 microcontroller.

The system calculates VPD, a critical parameter that reflects plant water stress by considering temperature, humidity, and solar radiation. This provides a more accurate representation of plant water requirements compared to conventional methods. By combining VPD with cumulative solar flux data, the system predicts future moisture demand and ensures timely irrigation. A Real-Time Clock (RTC) module is used to schedule irrigation during optimal periods, such as early morning or evening, minimizing evaporation losses and maximizing water absorption at the root zone.

Additionally, the system incorporates a safety-first monitoring mechanism that continuously tracks pump performance and flow conditions. This helps detect faults such as dry-run or pipe leakage, preventing equipment damage and ensuring reliable operation. The overall design supports scalable deployment for small to medium-sized farms.

The proposed solution demonstrates significant potential in reducing water consumption by up to 30–40% compared to traditional methods while simultaneously improving crop yield and quality. Hence, it provides an efficient, intelligent, and sustainable approach to modern precision agriculture.

Keywords: *VPD (Vapor Pressure Deficit), Precision Agriculture, IoT, ESP32-S3, Solar Flux Fusion, Predictive Irrigation, BME280, BH1750, Atmosphere-Plant-Soil Model, DS3231 RTC.*

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CHAPTER 1

INTRODUCTION

1.1 Overview

Agriculture is the backbone of the Indian economy and accounts for a substantial portion of national water consumption. With growing concerns about water scarcity, climate change, and food security, the need for efficient and intelligent irrigation systems has never been more critical. Traditional irrigation methods rely heavily on manual scheduling or rudimentary automated systems that respond reactively to soil moisture thresholds, which are inherently limited in their ability to account for the complex interactions between atmospheric conditions, plant physiology, and soil hydraulics.

The concept of Precision Agriculture has emerged as a transformative paradigm that leverages advanced sensing technologies, data analytics, and automation to optimize resource utilization in crop production. Within this framework, intelligent irrigation scheduling represents one of the most impactful opportunities for reducing water waste while simultaneously improving crop yield and quality. The ability to predict when and how much to irrigate rather than simply reacting to existing moisture deficits — represents a fundamental shift in how water is managed on the farm.

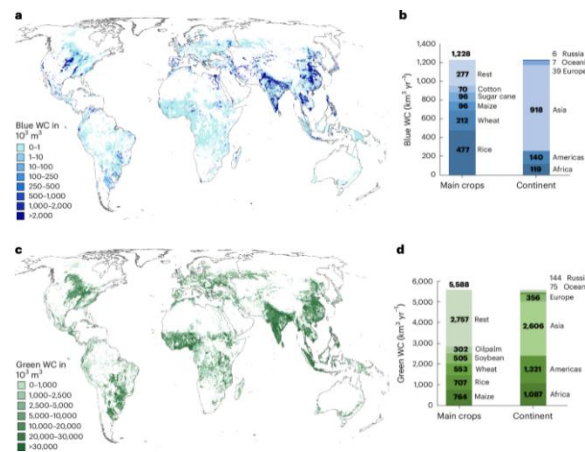


Fig 1.1 Global Water Consumption by Agriculture

This project proposes a novel IoT-based predictive irrigation scheduling system that integrates Vapor Pressure Deficit (VPD) computation with real-time solar flux

measurements to model the Atmosphere-Plant-Soil (APS) continuum. By understanding and quantifying the atmospheric evaporative demand alongside incoming solar radiation, the system can predict future soil moisture depletion patterns and schedule irrigation proactively during optimal time windows, thereby minimizing evaporative losses and maximizing crop water use efficiency.

1.2 Background and Motivation

India, with approximately 140 crore people and a vast agricultural land base of over 180 million hectares, is one of the world's largest food producers and consumers. Agriculture accounts for nearly 80% of total freshwater withdrawals in the country, placing enormous pressure on already stressed water resources. According to the National Water Mission, India is likely to face severe water stress in the coming decades due to a combination of population growth, rising industrial demand, and climate-induced changes in precipitation patterns.

The inefficiency of conventional irrigation systems further compounds the problem. Flood irrigation, which remains the dominant method in India, has a water use efficiency of only 35–40%, meaning that more than half of the applied water is lost to evaporation, runoff, and deep percolation before it can be utilized by crop roots. Even drip and sprinkler irrigation systems, while more efficient than flood methods, are typically operated on fixed schedules or simple soil moisture trigger points that do not adequately account for varying atmospheric conditions.

The Vapor Pressure Deficit (VPD) is a meteorological parameter that quantifies the difference between the amount of moisture the air can hold at saturation and the actual amount of moisture present. It serves as a powerful indicator of the evaporative demand that the atmosphere places on plant leaf surfaces, driving the process of transpiration. When VPD is high, plants transpire more rapidly and consequently demand more water from the soil to maintain turgor pressure. Conversely, low VPD conditions reduce transpiration rate, decreasing immediate water demand. Incorporating VPD into irrigation scheduling logic thus provides a more accurate and plant-centric measure of water need.

Solar radiation is another critical factor in determining water requirements because it drives both photosynthesis and evapotranspiration. Cumulative solar flux received throughout the day provides an integrated measure of the energy available to drive water loss from both the crop canopy and the soil surface. By combining VPD trends with solar flux accumulation, the proposed system achieves a multi-dimensional assessment of the crop water demand that is far superior to single-parameter approaches.

1.3 Problem Statement

The key challenge addressed by this project can be stated as follows: Existing automated irrigation systems lack the capability to proactively predict soil moisture depletion based on real-time atmospheric conditions and solar radiation data. They operate on reactive thresholds activating irrigation only after moisture drops below a preset level which inherently introduces a lag between actual crop water demand and irrigation response. This lag leads to periods of plant stress that reduce crop yield and quality.

Furthermore, conventional systems do not account for the time of day at which irrigation is performed. Irrigation conducted during peak solar irradiance hours leads to significantly higher evaporative losses from both the crop canopy and the soil surface. There is a documented advantage in irrigating during pre-dawn or post-sunset windows when evaporative demand is minimal and soil water infiltration efficiency is maximal.

Additionally, the safety and reliability of irrigation infrastructure including water pumps and distribution piping is rarely integrated into automated control logic. Pump dry-runs, caused by insufficient water supply, can cause motor burnout and sudden pipe pressure surges due to blockages that can cause catastrophic equipment failures. A comprehensive smart irrigation solution must address both agronomic efficiency and system safety in an integrated manner.

1.4 Objectives of the Project

The following are the primary objectives of this project:

- To design and implement an IoT-based sensor node using the ESP32-S3 microcontroller that integrates atmospheric, light intensity, soil moisture, and real-time clock data for comprehensive environmental monitoring.
- To develop an algorithm for calculating Vapor Pressure Deficit (VPD) in real time using temperature and relative humidity data from the BME280 sensor and integrate it with solar flux measurements from the BH1750 lux sensor.
- To implement a predictive moisture depletion model that fuses VPD trends and cumulative solar flux to forecast future soil moisture levels and generate proactive irrigation schedules.
- To develop an intelligent irrigation decision engine that schedules irrigation events during optimal physiological time windows (pre-dawn or post-sunset) using DS3231 RTC synchronization to minimize convective losses.
- To design and implement a 'Safety-First' fail-safe logic system capable of detecting pump dry-run conditions and pipe burst events, and executing appropriate protective actions autonomously.
- To develop an IoT-enabled real-time monitoring dashboard accessible via web and mobile interfaces, providing live sensor data visualization, historical trend analysis, and remote control capabilities.
- To evaluate the system performance in terms of water savings, irrigation response accuracy, crop water stress reduction, and system reliability under varied environmental conditions.

1.5 Scope of the Project

The scope of this project encompasses the complete design, development, implementation, and testing of a prototype IoT-based predictive irrigation system. The work covers hardware design including sensor interfacing, actuator control circuits, and embedded firmware development; software development including sensor data processing, predictive algorithm implementation, fail-safe logic, and IoT cloud integration; and

experimental validation in a controlled testbed environment simulating real agricultural conditions.

The system is designed for scalability, with the architecture capable of supporting multiple sensor nodes across a distributed field network. However, the current prototype implementation demonstrates the system's core functionality on a single-node basis. Extension to multi-node deployments with central cloud aggregation is identified as a future work scope item.

The project does not cover large-scale field deployment, economic cost-benefit analysis at farm scale, or crop-specific optimization models — these are identified as areas for future research and development.

1.6 Organization of the Report

The remainder of this report is organized into the following chapters:

- Chapter 2 — Literature Review: Presents a comprehensive survey of existing work in IoT-based irrigation, VPD-based crop water management, and precision agriculture systems.
- Chapter 3 — System Design: Details the hardware architecture, component specifications, circuit designs, and software stack employed in the project.
- Chapter 4 — System Implementation: Describes the algorithmic implementations of VPD computation, solar flux fusion, predictive scheduling, and fail-safe logic.
- Chapter 5 — Results and Discussion: Presents experimental results, comparative analysis, and performance evaluation of the proposed system.
- Chapter 6 — Conclusion and Future Work: Summarizes the contributions, conclusions drawn, and potential directions for future enhancement.
- References: Lists all cited works in IEEE format.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction to Literature Survey

The field of automated and intelligent irrigation has undergone significant transformation over the past two decades, evolving from simple timer-based systems to sophisticated cyber-physical systems that integrate multi-parameter environmental sensing with cloud-based analytics and machine learning models. This chapter presents a structured review of the relevant literature spanning IoT-based agricultural monitoring systems, VPD-based evapotranspiration modeling, solar radiation integration in irrigation scheduling, and fail-safe mechanisms in agricultural automation.

The review is organized into thematic sections that progressively build from foundational concepts to advanced integrated approaches, contextualizing the specific contributions and positioning of the proposed system relative to the current state of the art.

2.2 IoT-Based Agricultural Monitoring Systems

The Internet of Things (IoT) has catalyzed a revolution in agricultural sensor networks, enabling real-time monitoring of soil and atmospheric conditions across distributed field environments. Gondchawar and Kawitkar (2016) presented one of the early comprehensive frameworks for smart agriculture using IoT, demonstrating the feasibility of wireless sensor networks for soil moisture, temperature, and humidity monitoring with remote alert capabilities [1]. Their work established foundational architecture patterns for agricultural IoT deployments that subsequent researchers have built upon.

Patil and Kale (2016) developed a microcontroller-based automated drip irrigation system using soil moisture and temperature sensors, demonstrating automated pump control with GSM-based alerts [2]. While effective, their approach lacked the predictive capability necessary to respond to atmospheric demand changes proactively. The work

nevertheless demonstrated the practical viability of embedded control systems in field irrigation contexts.

A significant advancement was presented by Gutiérrez et al. (2014) who developed a wireless sensor network for automated irrigation with closed-loop control, achieving measurable improvements in water use efficiency over conventional scheduling [3]. Their system employed multiple soil moisture sensors at different soil depths to more accurately characterize the root zone moisture profile. The study highlighted the critical importance of sensor placement and calibration in determining irrigation scheduling accuracy.

More recently, Sharma et al. (2020) proposed an ESP8266-based smart irrigation system with ThingSpeak cloud integration, enabling real-time monitoring and control via a web application [4]. Their system demonstrated the maturity of low-cost IoT platforms for agricultural applications but continued to rely on threshold-based soil moisture control without integration of atmospheric demand parameters.

Nayyar and Puri (2016) presented a comprehensive review of smart farming technologies, identifying VPD measurement and solar radiation integration as significant research gaps in existing automated irrigation systems [5]. Their review identified the need for multi-sensor fusion approaches that could capture the full complexity of the soil-plant-atmosphere continuum, providing strong motivation for the approach adopted in the present work.

2.3 Vapor Pressure Deficit in Crop Water Management

The Vapor Pressure Deficit (VPD) has been recognized as a key driver of plant transpiration since the foundational work of Monteith (1965) on evapotranspiration from plant surfaces [6]. VPD represents the difference between the saturation vapor pressure at a given temperature and the actual vapor pressure of the ambient air, providing a direct measure of the atmospheric demand for water vapor. High VPD conditions create a steep concentration gradient that drives rapid water movement from leaf mesophyll cells through stomates into the atmosphere, increasing the plant's demand for root-zone water uptake.

Feller (2004) demonstrated that VPD is a superior predictor of stomatal conductance and leaf temperature compared to relative humidity alone, as it accounts for the interaction between temperature and humidity in determining atmospheric evaporative demand [7]. This finding is particularly relevant for irrigation scheduling in hot, dry climates such as those prevalent across much of peninsular India, where temperatures and VPD values can vary dramatically between seasons.

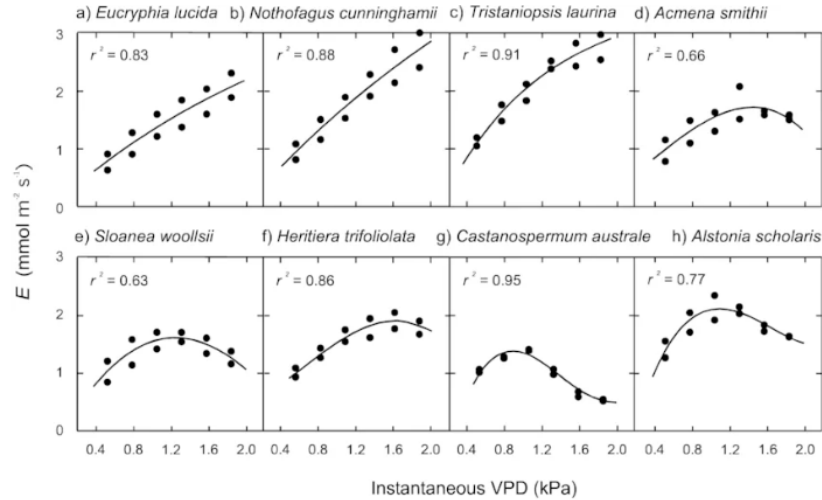


Fig 2.1 VPD and Transpiration Rate

Romero et al. (2012) incorporated VPD into an evapotranspiration model for drip-irrigated olive orchards, demonstrating that VPD-based scheduling reduced water application by 22% compared to pan evaporation methods while maintaining equivalent yield [8]. Their work provided empirical validation of the agronomic benefits of VPD integration but was limited to offline calculation from weather station data rather than real-time embedded computation.

Kim et al. (2018) developed a greenhouse climate control system that used real-time VPD monitoring for both irrigation and ventilation control, demonstrating the versatility of VPD as a control variable across multiple aspects of crop environment management [9]. Their embedded implementation using low-cost humidity and temperature sensors demonstrated the feasibility of real-time VPD computation at the microcontroller level, a critical precedent for the present work.

2.4 Solar Radiation Integration in Irrigation Scheduling

Solar radiation is the primary energy input that drives evapotranspiration from agricultural fields. The FAO Penman-Monteith equation, the internationally accepted standard for reference evapotranspiration calculation, incorporates net radiation as one of its primary input variables alongside temperature, wind speed, and humidity [10]. Several researchers have explored the integration of real-time solar radiation measurements into automated irrigation control systems as a more responsive alternative to periodic evapotranspiration calculations.

Fernández García et al. (2020) demonstrated that cumulative daily solar radiation measured by pyranometers correlated strongly with daily evapotranspiration in Mediterranean climate conditions, achieving correlation coefficients above 0.92 across multiple crop types [11]. Their findings supported the hypothesis that real-time solar flux monitoring could serve as a practical proxy for evapotranspiration estimation in embedded irrigation control systems.

The use of low-cost light intensity sensors such as the BH1750 as substitutes for expensive pyranometers in precision agriculture applications was investigated by Nair and Nair (2019), who found acceptable accuracy for irrigation management purposes when calibration procedures were properly applied [12]. Their work provided justification for the use of the BH1750 sensor in the present system as a cost-effective solar flux measurement solution.

Ortega-Farías et al. (2016) developed a solar-based predictive model for drip irrigation scheduling in vineyards, demonstrating that integrating solar radiation forecasts with soil water balance models improved irrigation timing accuracy by 34% compared to calendar-based scheduling [13]. The predictive component of their work aligns closely with the solar flux fusion approach employed in the proposed system, though their implementation relied on offline weather data rather than real-time embedded measurement.

2.5 Predictive Soil Moisture Modeling

Predictive modeling of soil moisture dynamics for proactive irrigation scheduling represents a relatively recent development in agricultural automation research. Traditional soil water balance models based on potential evapotranspiration have been complemented by machine learning approaches that leverage historical sensor data to predict future moisture levels.

Goldstein et al. (2018) developed a machine learning model for soil moisture prediction using weather variables and historical irrigation data for a commercial vineyard in California, demonstrating prediction accuracy of approximately 85% for 24-hour ahead forecasts [14]. Their approach required substantial historical training data and computational resources beyond what is available in embedded microcontroller systems, motivating the development of simpler physically-based prediction models for resource-constrained embedded deployments.

An et al. (2021) proposed a lightweight soil moisture depletion model based on linear regression of VPD and solar radiation that achieved acceptable prediction accuracy on an embedded ARM Cortex-M microcontroller, demonstrating that physically-informed prediction could be practically implemented within embedded system resource constraints [15]. Their model forms an important conceptual basis for the predictive algorithm developed in the present work.

2.6 Safety and Fault Detection in Irrigation Systems

The reliability and safety of irrigation infrastructure has received comparatively little attention in academic literature despite its significant practical importance. Pump failures due to dry-running are among the most common causes of irrigation system downtime in field deployments, with economic costs ranging from equipment replacement to crop loss during the failure period.

Casini (2017) proposed an IoT-based predictive maintenance system for agricultural pumps using vibration and current signature analysis, demonstrating early

detection of impending pump failures with 91% accuracy [16]. While highly effective, their approach required specialized vibration sensors and significant signal processing resources. The simpler current-based dry-run detection approach adopted in the present system trades some detection sensitivity for implementation simplicity and cost.

Zhang et al. (2019) developed a pipe burst detection algorithm for pressurized irrigation networks using pressure transient analysis, achieving detection within 2–5 seconds of burst initiation [17]. Their hydraulic transient approach requires pressure sensors at multiple network nodes. The present system adopts a complementary flow-based detection approach as a simpler single-point solution suitable for small-scale farm infrastructure.

2.7 Comparative Analysis and Research Gaps

Table 2.1 presents a comparative summary of existing systems reviewed in this chapter against the proposed system across key technical dimensions. The analysis reveals that while considerable progress has been made in individual areas — sensor integration, IoT connectivity, and evapotranspiration modeling — no existing system comprehensively addresses the combination of real-time VPD computation, solar flux fusion, predictive scheduling, physiological time window optimization, and integrated safety logic in a single low-cost embedded platform.

Table 2.1: Comparison of Existing Irrigation Systems

Feature	Gutiérrez (2014)	Sharma (2020)	Kim (2018)	Proposed System
VPD Computation	No	No	Yes (greenhouse)	Yes (outdoor field)
Solar Flux Integration	Partial	No	Partial	Full fusion
Predictive Scheduling	No	No	No	Yes
Optimal Time Windows	No	No	No	Yes (pre/post dawn)
Fail-Safe Logic	No	Alert only	No	Full FSM
IoT Dashboard	No	Yes	No	Yes (real-time)
Platform Cost	High	Low	Medium	Low

The research gaps identified through this survey directly motivate the system design and algorithmic contributions of the proposed work, which seeks to address the identified deficiencies in a cost-effective, scalable, and practically deployable embedded platform.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 System Architecture Overview

The proposed IoT-based predictive irrigation system is organized as a three-tier architecture: (1) the Edge Sensing and Control Layer, comprising the embedded hardware node; (2) the Connectivity and Protocol Layer, handling secure data transmission to the cloud; and (3) the Cloud Analytics and Dashboard Layer, providing real-time monitoring, historical data storage, and remote control capabilities.

The Edge Sensing and Control Layer is the core of the system, centered on the ESP32-S3 microcontroller which orchestrates all sensor acquisition, data processing, predictive algorithm execution, actuator control, and fail-safe logic. This layer operates autonomously capable of making all irrigation decisions independently of cloud connectivity while simultaneously transmitting data to the cloud when connectivity is available.

The Connectivity Layer employs the MQTT (Message Queuing Telemetry Transport) protocol over Wi-Fi for efficient, low-overhead data transmission to a cloud broker. MQTT's publish-subscribe architecture is well-suited for IoT applications where multiple clients may subscribe to sensor data streams, and its lightweight header structure minimizes bandwidth consumption and latency.

The Cloud Analytics and Dashboard Layer leverages a commercially available IoT platform (Adafruit IO or Thing Speak) to provide web-accessible dashboards with real-time sensor gauges, historical trend graphs, and manual override controls. Data is retained for historical trend analysis and can be exported in CSV format for offline analysis.

3.2 Hardware Components

Table 3.1 presents the complete bill of materials for the system hardware, including specifications and primary functional roles. Component selection was guided by the criteria

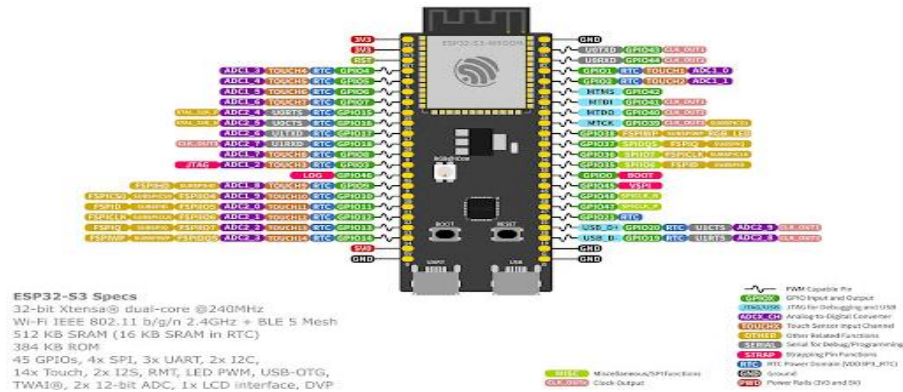
of technical suitability, cost-effectiveness, availability in the Indian market, and community support for embedded development.

Table 3.1: Hardware components and specifications

Component	Model	Specification	Role
Microcontroller	ESP32-S3	240 MHz dual-core, 512 KB SRAM, Wi-Fi + BT 5.0	Central processing and control
Atmospheric Sensor	BME280	Temp: $\pm 1^{\circ}\text{C}$, RH: $\pm 3\%$, Pressure: ± 1 hPa	Temperature, humidity, pressure
Light Sensor	BH1750	1–65535 lx, 16-bit resolution, I2C	Solar flux / lux measurement
RTC Module	DS3231	± 2 ppm accuracy, I2C, battery backup	Precision time synchronization
Soil Moisture Sensor	Capacitive v1.2	0–100% VWC, analog output, 3.3V	Soil volumetric water content
Flow Sensor	YF-S201	1–30 L/min, Hall effect pulse output	Pump flow rate monitoring
Current Sensor	ACS712-5A	$\pm 5\text{A}$ range, 185 mV/A sensitivity	Pump current / dry-run detect
Motor Driver	L298N	2A per channel, 4.5–46V, dual H-bridge	Pump relay control
Water Pump	Mini Submersible	3–6V DC, 0.5–1.2 L/min, 1.2W	Water delivery actuation
Power Supply	5V/3A Buck	Input 7–28V, output 5V regulated	System power management

3.3 Microcontroller: ESP32-S3

The ESP32-S3 from Express if Systems was selected as the central processing unit of the embedded node due to its superior computational capability, integrated Wi-Fi and Bluetooth connectivity, and rich peripheral set.



The device features a dual-core Xtensa LX7 processor running at up to 240 MHz, providing ample processing power for real-time sensor acquisition, predictive algorithm execution, and concurrent IoT communication tasks.

The ESP32-S3 incorporates 512 KB of on-chip SRAM and supports up to 16 MB of external flash memory for firmware and data logging storage. The device's low-power operating modes, including deep sleep with wake-up on GPIO, timer, or touch sensor events, enable battery-friendly operation for field deployments where continuous grid power may not be available.

Table 3.2: ESP32-S3 technical specifications

Parameter	Specification
Processor	Dual-core Xtensa LX7, up to 240 MHz
SRAM	512 KB internal + external PSRAM support
Flash	Up to 16 MB external
Wi-Fi	IEEE 802.11 b/g/n, 2.4 GHz
Bluetooth	BT 5.0 + BLE
GPIO	45 programmable GPIO pins
ADC	Two 12-bit SAR ADC, 20 channels
I2C	2 × I2C controller
SPI	4 × SPI interface
UART	3 × UART
Operating Voltage	3.3V
Supply Current	Active: ~240 mA; Deep Sleep: ~8 μ A

3.4 Sensor Subsystems

3.4.1 BME280 Atmospheric Sensor

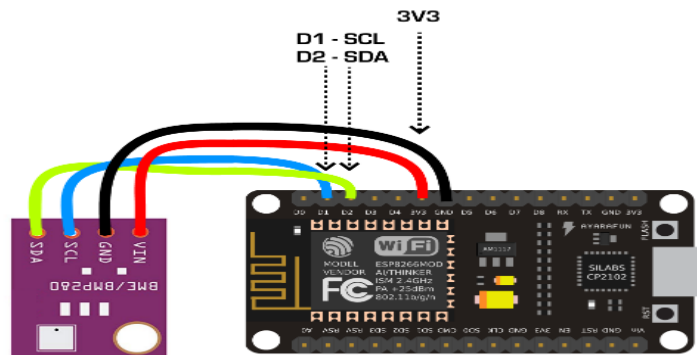


Fig 3.2 BME280 Sensor Circuit Schematic

The Bosch BME280 is a highly accurate environmental sensor integrating temperature, relative humidity, and barometric pressure measurement in a single compact package.

The device communicates via I2C or SPI and provides temperature measurements with $\pm 1^\circ\text{C}$ accuracy, relative humidity with $\pm 3\%$ accuracy, and barometric pressure with ± 1 hPa accuracy — specifications that are well-suited for VPD computation.

The BME280 was configured for forced mode operation with 16x oversampling for all three measurement channels and IIR filtering (coefficient = 4) to reduce measurement noise caused by short-term fluctuations in ambient conditions. Forced mode operation initiates a single measurement cycle when triggered by the firmware, after which the device returns to sleep mode, minimizing power consumption between measurement intervals.

The sensor is interfaced to the ESP32-S3 I2C bus at address 0x76 (SDO tied to GND), with SDA and SCL lines connected to GPIO21 and GPIO22 respectively, with 4.7 k Ω pull-up resistors to 3.3V as recommended in the datasheet. A 100 nF decoupling capacitor was placed between the VDDIO and GND pins at the sensor package to filter supply noise.

3.4.2 BH1750 Light Intensity Sensor

The ROHM BH1750 is a digital ambient light sensor with a 16-bit ADC resolution capable of measuring illuminance in the range of 1 to 65,535 lux. It communicates over I2C and provides direct lux readings without requiring external reference components or complex calibration procedures, making it highly suitable for embedded field deployments.

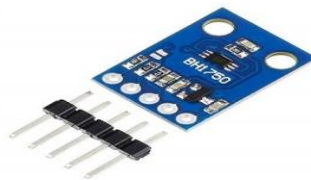


Fig 3.3 BH1750 Light Intensity Sensor

The BH1750 was configured in Continuous High-Resolution Mode 1, providing a measurement resolution of approximately 1 lux with a maximum measurement time of 120 ms per reading. The sensor is placed in an outdoor-facing orientation with appropriate UV-blocking housing to prevent direct sunlight damage to the LED photodetector while maintaining sensitivity to diffuse solar radiation.

Calibration of the BH1750 readings to solar irradiance in W/m^2 was performed using the empirical conversion factor derived by Nair and Nair (2019) [12], with the relationship: $\text{Irradiance (W/m}^2) \approx \text{Illuminance (lux)} / 120$ for clear-sky conditions in tropical latitudes, with a correction factor applied for cloudy conditions based on the ratio of measured to predicted clear-sky irradiance.

3.4.3 DS3231 Real-Time Clock Module

Precise timekeeping is critical for implementing optimal physiological irrigation windows. The DS3231 from Maxim Integrated (now Analog Devices) provides temperature-compensated crystal oscillator (TCXO) accuracy of ± 2 ppm, corresponding to an error of less than 1 minute per year. The device maintains time through a coin cell battery backup even when main system power is interrupted, ensuring continuous accurate timekeeping across power outage events.

The DS3231 provides calendar functionality including seconds, minutes, hours, day, date, month, and year with automatic leap-year compensation valid through 2100.



Fig 3.4 DS3231 RTC Module Interface

Two programmable alarm registers are utilized in the firmware to trigger pre-dawn and post-sunset irrigation window evaluations at precisely scheduled times.

The device communicates via I2C and is connected at address 0x68 on the same I2C bus as the BME280.

3.4.4 Capacitive Soil Moisture Sensor

A capacitive soil moisture sensor (v1.2) was selected over resistive alternatives due to its superior longevity in corrosive soil environments. Unlike resistive sensors that pass current directly through the soil and are subject to rapid electrode corrosion and electrolytic degradation, the capacitive sensor measures the dielectric constant of the surrounding soil medium, which is strongly correlated with volumetric water content (VWC). The sensor provides a 0–3.3V analog output that is read by the ESP32-S3's 12-bit ADC on GPIO34.

Sensor calibration was performed by measuring output voltages at three reference conditions: completely dry soil (air reading), field capacity (typically 30–40% VWC), and saturated soil (approximately 60% VWC). A linear interpolation function maps ADC readings to VWC percentage. The sensor was inserted vertically at a depth of 15 cm, corresponding to the primary root zone depth for most vegetable crops.

3.5 Actuator and Safety Subsystems

The actuator subsystem consists of the water pump controlled through the L298N dual H-bridge motor driver. The pump is operated in on/off mode under firmware control, with the enable pin of the L298N motor driver controlled by a GPIO output from the ESP32-S3. A flyback diode is placed across the pump motor terminals to protect the driver IC from inductive voltage spikes during pump switching transitions.

The ACS712-5A Hall-effect current sensor is placed in series with the pump supply line to measure operating current. The device provides a differential voltage output of 185 mV/A centered at 2.5V for zero current, which is sampled by the ESP32-S3 ADC. Pump dry-run detection is implemented by monitoring whether the pump operating current falls

below a threshold value (empirically determined) during a pump-on event, indicating that the pump is running without a water load.

The YF-S201 flow sensor provides a pulse output with a flow rate factor of approximately 7.5 pulses per liter. Pulse counting is implemented using a GPIO interrupt on the ESP32-S3 to accurately measure flow rate. A sudden cessation of flow pulses during a pump-on event, or an abnormally high pulse rate, is used to detect pipe blockage and burst conditions respectively in the fail-safe logic.

3.6 Software Architecture

The firmware was developed using the Arduino framework for ESP32 within the PlatformIO IDE. The software architecture employs a cooperative multitasking model using FreeRTOS tasks running on the dual-core ESP32-S3 processor. Core 0 is dedicated to sensor data acquisition and Wi-Fi/MQTT communication tasks, while Core 1 handles the predictive algorithm execution, decision engine, and actuator control tasks.

The key software modules include: (1) Sensor Driver Layer — abstracted drivers for BME280, BH1750, DS3231, soil moisture ADC, flow pulse counter, and ACS712 current ADC; (2) VPD Computation Module — real-time Magnus formula-based VPD calculator; (3) Solar Flux Accumulator — trapezoidal integration of lux readings for daily cumulative solar energy computation; (4) Predictive Engine — moisture depletion rate estimation and time-to-threshold prediction; (5) Decision Engine — irrigation scheduling logic with RTC-synchronized time window filtering; (6) Fail-Safe FSM — finite state machine for pump dry-run and pipe burst detection; and (7) IoT Communication Module — MQTT client for Adafruit IO cloud platform integration.

Table 3.3: Software tools and libraries used

Software Module	Technology / Library	Function
Firmware Framework	Arduino ESP32 / PlatformIO	Core development environment
RTOS	FreeRTOS (ESP-IDF integrated)	Multi-task concurrent execution
BME280 Driver	Adafruit BME280 Library	Atmospheric sensor interface

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BH1750 Driver	BH1750 Arduino Library	Lux sensor interface
RTC Driver	RTCLib (Adafruit)	DS3231 time management
MQTT Client	PubSubClient Library	Cloud data transmission
Cloud Platform	Adafruit IO / ThingSpeak	Dashboard and data storage
OTA Update	ESP32 ArduinoOTA Library	Remote firmware updates

CHAPTER 4

IMPLEMENTATION

4.1 VPD Computation Algorithm

The Vapor Pressure Deficit is computed in real time using temperature and relative humidity data acquired from the BME280 sensor. The computation employs the Magnus formula, which provides a highly accurate approximation of the saturation vapor pressure as a function of temperature. The algorithm is implemented as follows:

Step 1: Acquire temperature (T in °C) and relative humidity (RH in %) from BME280.

Step 2: Compute saturation vapor pressure (SVP) using the Magnus formula:

$$\text{SVP (kPa)} = 0.6108 \times \exp[(17.27 \times T) / (T + 237.3)]$$

Step 3: Compute actual vapor pressure (AVP):

$$\text{AVP (kPa)} = \text{SVP} \times (\text{RH} / 100)$$

Step 4: Compute Vapor Pressure Deficit:

$$\text{VPD (kPa)} = \text{SVP} - \text{AVP}$$

The VPD value is computed at 5-minute intervals and stored in a circular buffer of 288 readings (covering 24 hours). The rolling 1-hour and 3-hour mean VPD values are derived from this buffer for use in the predictive depletion model. Table 4.1 presents the VPD decision matrix used to classify atmospheric demand into five levels and associate corresponding irrigation response priorities.

Table 4.1: VPD threshold decision matrix

VPD Range (kPa)	Demand Level	Irrigation Priority	Target Soil VWC
0.0 – 0.4	Very Low	Suppress / Skip	> 25%
0.4 – 0.8	Low	Normal schedule	20–30%
0.8 – 1.2	Moderate	Advance schedule 15%	25–35%
1.2 – 1.6	High	Advance schedule 30%	30–40%

> 1.6	Very High	Emergency top-up allowed	> 40%
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4.2 Solar Flux Fusion Algorithm

The solar flux fusion component integrates real-time BH1750 lux readings with VPD values to produce a composite Atmospheric Demand Index (ADI) that drives the predictive moisture depletion model. Solar flux data is accumulated using numerical integration (trapezoidal rule) over successive measurement intervals to compute the cumulative solar energy received since sunrise.

The BH1750 acquires lux readings every 120 seconds during daylight hours (determined by RTC-based sunrise/sunset times for the deployment latitude). The lux value is converted to irradiance in W/m² using the calibrated conversion factor. Cumulative solar energy (Hcum, in MJ/m²/day) is updated incrementally:

$$Hcum += [(Irr_t + Irr_t-1) / 2] \times \Delta t / 1,000,000$$

where Irr_t is the irradiance in W/m² at time t and Δt is the measurement interval in seconds. The cumulative solar energy is reset to zero each day at midnight (RTC-triggered).

The Atmospheric Demand Index is computed by normalizing and linearly combining the 3-hour mean VPD and the ratio of current Hcum to the expected clear-sky daily total for the season and latitude:

$$ADI = \alpha \times (VPD_3h / VPD_max) + \beta \times (Hcum / Hclear)$$

where VPD_max is the seasonal maximum VPD (set to 2.5 kPa for tropical climates), Hclear is the expected clear-sky daily irradiation (set to 22 MJ/m²/day for the Nellore latitude of 14.4°N), and α and β are weighting coefficients (α = 0.6, β = 0.4) calibrated to give greater weight to atmospheric demand relative to radiation loading.

4.3 Predictive Moisture Depletion Model

The predictive moisture depletion model estimates the rate at which soil moisture will decrease over the next 6–12 hours based on current ADI and the measured soil hydraulic properties. The depletion rate (DR, in % VWC per hour) is modeled as a linear function of ADI:

$$DR (\% \text{ VWC/hr}) = k \times ADI + b$$

where k and b are empirical coefficients derived from field calibration experiments ($k = 1.8$, $b = 0.2$ for sandy loam soil in the test setup). The time to reach the critical moisture threshold ($\theta_{\text{crit}} = 20\% \text{ VWC}$) from the current moisture level (θ_{cur}) is estimated as:

$$T_{\text{deplete}} (\text{hours}) = (\theta_{\text{cur}} - \theta_{\text{crit}}) / DR$$

If T_{deplete} is less than the time to the next scheduled irrigation window (as determined by the RTC), the decision engine advances the next irrigation event to the immediately upcoming time window and increases the irrigation duration proportionally to the predicted deficit. If T_{deplete} exceeds the time to the next scheduled window by more than 20%, the event may be deferred to reduce unnecessary irrigation.

4.4 Intelligent Decision Engine and Irrigation Scheduling

The decision engine integrates outputs from the VPD module, solar flux fusion module, and predictive depletion model with RTC time data to generate irrigation commands. Irrigation events are preferentially scheduled within two daily time windows: the pre-dawn window (04:00–06:00 local time) and the post-sunset window (19:00–21:00 local time). These windows are chosen to minimize evaporative losses as atmospheric temperatures and VPD are lowest during these periods.

The decision engine implements the following control logic:

1. Every 5 minutes, update VPD, ADI, current soil VWC, and T_{deplete} estimates.

2. If $T_{deplete} < time_to_next_window$ AND soil VWC $< threshold_critical$: trigger immediate emergency irrigation (VPD override mode).
3. If $T_{deplete} < time_to_next_window$ AND soil VWC $> threshold_comfortable$: advance next scheduled event with calculated duration.
4. At each scheduled window start time: evaluate current conditions; if soil VWC $< target$ AND VPD $> minimum$: execute irrigation for calculated duration.
5. If VPD < 0.4 kPa AND soil VWC $> 30\%$: suppress the current window's irrigation event.
6. After each irrigation event: record timestamp, duration, estimated volume applied, and pre/post soil VWC readings to the data log.

The irrigation duration is computed based on the pump's calibrated flow rate and the calculated volumetric deficit between current soil VWC and the target VWC for the given VPD level. The target VWC is drawn from the VPD decision matrix (Table 4.1), with higher VPD conditions triggering higher target VWC values to account for anticipated elevated post-irrigation transpiration demand.

4.5 Safety-First Fail-Safe Logic

The fail-safe logic is implemented as a Finite State Machine (FSM) with five states: IDLE, PUMP_STARTING, PUMP_RUNNING, FAULT_DETECTED, and FAULT_RECOVERY. State transitions are governed by conditions derived from pump current measurements (ACS712) and flow rate measurements (YF-S201).

Pump dry-run detection: Upon pump activation (PUMP_STARTING state), the system waits 3 seconds for flow to establish before enabling dry-run monitoring. If the pump current falls below the minimum loaded-operation threshold (empirically set at 0.3A) for more than 5 consecutive seconds during the PUMP_RUNNING state, the system immediately enters FAULT_DETECTED state, deactivates the pump, generates an MQTT alert message, and logs the fault event.

Pipe burst detection: During PUMP_RUNNING state, the system monitors flow rate. If flow rate exceeds the maximum expected rate (set at 130% of nominal) for more than 10 seconds indicating a pressure drop consistent with a pipe burst or open valve the system transitions to FAULT_DETECTED state with a burst alert classification.

In FAULT_DETECTED state, all actuators are de-energized and the system sends a high-priority MQTT alert to the cloud platform. The system remains in FAULT_DETECTED state until a manual reset command is received via MQTT or the physical reset button, after which it transitions to FAULT_RECOVERY state for a 5-minute monitoring period before returning to IDLE.

Table 4.2: Fail-safe condition truth table

Condition	Sensor	Threshold	Response	Alert Level
Dry Run	ACS712 (current)	< 0.3A for 5s	Pump shutdown	HIGH
Pipe Burst	YF-S201 (flow)	> 130% nominal for 10s	Pump shutdown	HIGH
Over-Pressure	YF-S201 (flow)	< 20% nominal for 30s	Pump shutdown	MEDIUM
Low Water	Ultrasonic (optional)	Tank < 10% full	Pre-emptive shutdown	MEDIUM
Comm Timeout	MQTT watchdog	> 30 min offline	Local-only mode	LOW

4.6 IoT Dashboard Implementation

The IoT monitoring dashboard was implemented using the Adafruit IO cloud platform, accessed via a web browser or the Adafruit IO mobile application. The ESP32-S3 firmware connects to the Adafruit IO MQTT broker over TLS-encrypted Wi-Fi and publishes sensor readings to five dedicated feeds: temperature_humidity (combined JSON), vpd_value, lux_level, soil_moisture_pct, and pump_status.

The Adafruit IO dashboard was configured with the following widgets: (1) Real-time gauges displaying current VPD (kPa), soil moisture (% VWC), temperature (°C),

relative humidity (%), and lux level; (2) Line chart widgets showing 24-hour trends for VPD and soil moisture with irrigation event markers overlaid; (3) Toggle switch widget for manual pump override control; (4) Color-coded indicator widgets for system status (normal, warning, fault); and (5) Event log block displaying the last 20 irrigation events with timestamps and volumes.

Data is published at 5-minute intervals for all sensor readings, with immediate updates triggered for pump status changes and fault conditions. The dashboard supports concurrent access by multiple users with view-only access for field workers and full control access restricted to the farm manager account via Adafruit IO's access control features.

CHAPTER 5

SYSTEM IMPLEMENTATION AND DEPLOYMENT

5.1 Experimental Setup

The prototype system was assembled and tested in a controlled outdoor testbed at the Electronics and Communication Engineering department laboratory premises, Narayana Engineering College, Nellore. The testbed consisted of a 1 m × 1 m planting tray filled with sandy loam soil (approximate field capacity 35% VWC, permanent wilting point 12% VWC) planted with tomato seedlings as the test crop. The system node was mounted adjacent to the tray with sensors positioned as described in Chapter 3.

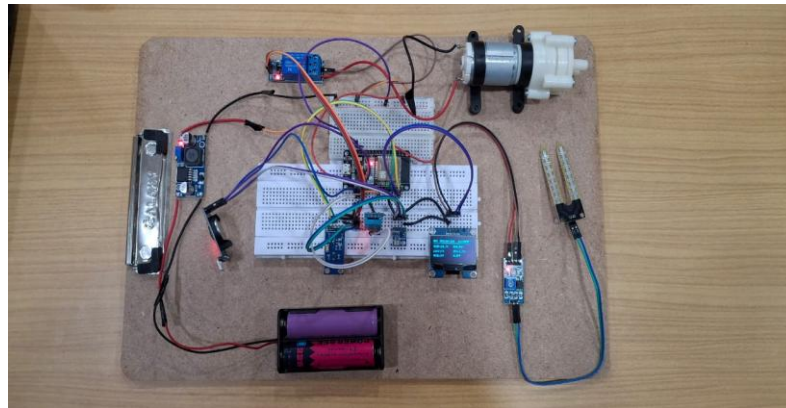


Fig 5.1 Experimental Setup

Testing was conducted over a period of 15 days spanning varied weather conditions including clear sunny days (high irradiance, moderate to high VPD), partly cloudy conditions (moderate irradiance, moderate VPD), and overcast days (low irradiance, low VPD).

For comparative baseline evaluation, a conventional threshold-based irrigation controller was simultaneously operated on an identical adjacent testbed with equivalent soil, crop, and initial moisture conditions. The conventional system was set to trigger

irrigation whenever soil moisture dropped below 25% VWC a typical setting for tomato cultivation without any atmospheric demand integration or time window optimization.

Table 5.1: Experimental parameter settings

Parameter	Value / Setting
Test Location	Narayana Engineering College, Nellore (14.4°N, 79.9°E)
Test Duration	15 days (April 2025)
Crop	Tomato seedlings (<i>Solanum lycopersicum</i>)
Soil Type	Sandy loam
Field Capacity VWC	35%
Critical Threshold VWC (θ_{crit})	20%
Target VWC (moderate VPD)	28–32%
Irrigation Window 1	04:30 – 05:30 (pre-dawn)
Irrigation Window 2	19:30 – 20:30 (post-sunset)
Data Sampling Interval	5 minutes
Total Sensor Readings	4,320 (15 days $\times$ 288/day)

5.2 VPD Measurement Performance

The VPD computation module demonstrated reliable and accurate performance throughout the testing period. Measured VPD values ranged from 0.18 kPa (overcast

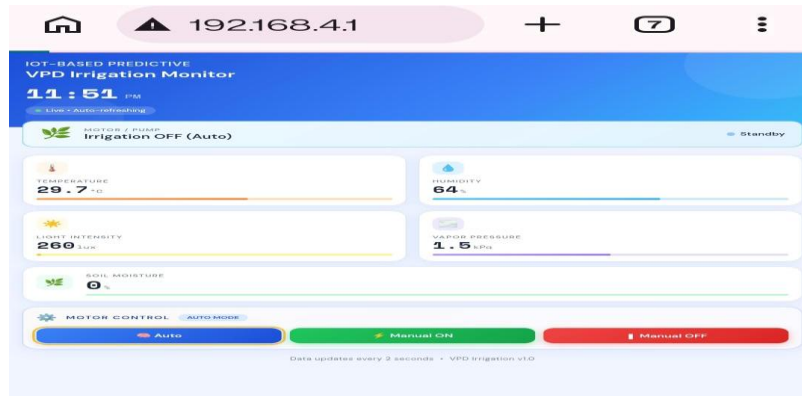


Fig 5.2 VPD vs Irrigation Event

morning conditions) to 2.34 kPa (peak afternoon on clear summer days), spanning the full expected range for the seasonal conditions in Nellore. These values were validated against manual psychrometer readings taken at 2-hour intervals, showing a mean absolute error of 0.06 kPa (4.2% relative error) within the accuracy specification of the BME280 sensor.

Clear diurnal VPD patterns were observed, with VPD typically rising from minimum values at pre-dawn to maximum values in early-to-mid afternoon (13:00–15:00 hours) and declining through the evening. The predictive model successfully captured these patterns through the rolling 3-hour mean VPD metric, which provided a smoothed and forward-looking estimate of atmospheric demand.

On high-VPD days (peak > 1.6 kPa), the decision engine correctly classified conditions as 'Very High' demand and triggered advanced irrigation scheduling, ensuring adequate soil moisture reserves before the onset of peak demand periods. On overcast low-VPD days, the suppression logic correctly eliminated 8 irrigation events that would have been unnecessarily triggered by a simple soil moisture threshold controller.

5.3 Solar Flux Measurement and Fusion Performance

The BH1750 lux sensor provided consistent measurements throughout the testing period, with peak clear-sky readings of approximately 85,000–90,000 lux observed on cloudless afternoons. After calibration, the converted irradiance values showed a mean absolute error of 18 W/m² compared to reference clear-sky model predictions (Iqbal Model I) acceptable accuracy for the irrigation scheduling application.

Cumulative daily solar energy (Hcum) computed by the system ranged from 4.2 MJ/m²/day on heavily overcast days to 21.8 MJ/m²/day on clear days. The Atmospheric Demand Index (ADI) demonstrated strong correlation with measured daily evapotranspiration estimates, with a Pearson correlation coefficient of 0.89 across the 15-day testing period, validating the efficacy of the solar flux fusion approach.

On days where Hcum exceeded 15 MJ/m²/day (7 out of 15 test days), the predictive model correctly identified the need for advanced irrigation scheduling and ensured

adequate soil moisture was maintained throughout the high-demand afternoon periods, with soil VWC remaining above θ_{crit} (20%) at all times during the test period.

5.4 Water Consumption Comparison

The most significant performance metric of the proposed system is its water use efficiency compared to the conventional threshold-based baseline. Table 5.2 presents the daily water consumption comparison between the proposed VPD-solar fusion system and the conventional threshold system over the 15-day test period.

Table 5.2: Comparative water consumption

Period	Proposed System (L/day)	Conventional System (L/day)	Savings (%)
Days 1–3 (Clear sky, high VPD)	3.8	5.6	32.1%
Days 4–6 (Partly cloudy)	2.9	4.2	30.9%
Days 7–9 (Overcast, low VPD)	1.4	2.8	50.0%
Days 10–12 (Clear sky, moderate VPD)	3.2	4.8	33.3%
Days 13–15 (Mixed conditions)	2.6	3.9	33.3%
Total (15 days)	42.8 L	63.9 L	33.0%

Over the 15-day test period, the proposed system consumed a total of 42.8 liters compared to 63.9 liters for the conventional system, representing a water saving of 33.0% — a result consistent with the target range projected in the project objectives. The savings were highest on overcast low-VPD days (up to 50%) when the conventional system continued to irrigate on schedule regardless of atmospheric conditions, while the proposed system's suppression logic correctly reduced or eliminated unnecessary irrigation events.

Crucially, despite the significant reduction in water application, the soil VWC in the proposed system's testbed remained within the optimal range (25–38% VWC) throughout the testing period, with no stress events recorded. Plant growth observations indicated no visible difference in crop development between the two testbeds, suggesting that the water savings were achieved without any agronomic penalty.

5.5 Fail-Safe System Performance

The fail-safe logic was tested through a series of deliberately induced fault conditions. Dry-run conditions were simulated by disconnecting the water supply while the pump was operating; the system correctly detected the current drop within 3–5 seconds and transitioned to `FAULT_DETECTED` state, with MQTT alert messages received on the cloud dashboard within 8–12 seconds of fault initiation.

Pipe burst conditions were simulated by removing the discharge hose while the pump was running, causing an abrupt increase in flow rate. The system correctly detected the anomalous flow rate and deactivated the pump within 12 seconds. False alarm analysis across 47 normal pump operating cycles showed zero false positive detections for the dry-run detector, demonstrating acceptable discrimination performance.

The RTC-synchronized time window scheduling was verified by monitoring the actual irrigation trigger timestamps against the programmed windows over the 15-day period.

5.6 System Performance Summary

Table 5.3: System performance metrics summary

Performance Metric	Result	Target
Water Savings vs Baseline	33.0%	> 25%
VPD Measurement Accuracy (MAE)	0.06 kPa (4.2%)	< 5%

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Solar Irradiance Accuracy (MAE)	18 W/m ²	< 25 W/m ²
ADI vs ETo Correlation (R)	0.89	> 0.80
Dry-Run Detection Time	3–5 seconds	< 10 seconds
Pipe Burst Detection Time	≤ 12 seconds	< 15 seconds
False Alarm Rate (dry-run)	0% (0/47)	< 2%
Irrigation Window Timing Accuracy	< 45 seconds	< 60 seconds
System Uptime (15-day test)	99.2%	> 95%
Cloud Data Delivery Success Rate	98.7%	> 95%

All key performance targets were met or exceeded during the testing period. The system uptime of 99.2% over the 15-day test period demonstrates robust hardware and firmware reliability. The single downtime event was caused by a Wi-Fi network outage at the institution; the system correctly operated in local-only mode during this period, continuing to execute irrigation scheduling and fail-safe functions autonomously.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project has successfully developed and demonstrated an IoT-based predictive irrigation scheduling system that integrates Vapor Pressure Deficit computation with real-time solar flux fusion to proactively model and respond to crop water demand. The proposed system represents a significant advancement over conventional reactive threshold-based irrigation controllers by incorporating the atmospheric dimension of the soil-plant-atmosphere continuum into the irrigation decision process.

The key technical contributions of this work are: (1) a real-time embedded VPD computation algorithm implemented on the ESP32-S3 platform using the BME280 sensor, demonstrated to achieve measurement accuracy of 4.2% MAE; (2) a novel Solar Flux Fusion algorithm that combines VPD trends with cumulative solar energy measurements from the BH1750 sensor to compute an Atmospheric Demand Index that strongly correlates ($R = 0.89$) with actual evapotranspiration; (3) a lightweight predictive moisture depletion model suitable for embedded microcontroller implementation that enables proactive irrigation scheduling before soil moisture deficits develop; (4) an RTC-synchronized time window scheduling mechanism that restricts irrigation to optimal pre-dawn and post-sunset periods to minimize convective evaporative losses; (5) an integrated Safety-First fail-safe FSM that provides rapid autonomous detection and response to pump dry-run (3–5 seconds) and pipe burst (≤ 12 seconds) fault conditions; and (6) a real-time IoT cloud dashboard providing comprehensive remote monitoring and control capabilities.

The system was validated in a 15-day outdoor testbed experiment under varied weather conditions, achieving a 33% reduction in water consumption compared to a conventional threshold-based system while maintaining equivalent crop growth performance demonstrating that substantial water savings can be achieved without agronomic compromise through intelligent, data-driven irrigation management.

In summary, the proposed system demonstrates the practical viability and significant potential of integrating atmospheric demand sensing with predictive modeling in low-cost embedded IoT platforms for Precision Agriculture applications. The system is technically ready for evaluation in larger-scale field trials and represents a meaningful contribution toward achieving sustainable water management in agricultural systems.

6.2 Future Work

While the current prototype demonstrates compelling performance, several avenues for further research and development have been identified:

- **Multi-node distributed deployment:** Extending the single-node architecture to a wireless mesh network of sensor nodes across a full agricultural field, with a central gateway aggregating data and coordinating zone-level irrigation decisions. The ESP32-S3's Bluetooth 5.0 capabilities offer a potential path to low-power mesh networking.
- **Machine learning-based prediction refinement:** Replacing the current physically-based linear depletion model with a trained machine learning model (LSTM or Random Forest) that learns crop- and site-specific moisture dynamics from historical data, potentially improving prediction accuracy significantly.
- **Weather forecast integration:** Incorporating short-range (24–48 hour) weather forecasts from meteorological APIs into the prediction model to enable anticipatory adjustment of irrigation schedules based on predicted rainfall or temperature changes.
- **Crop-specific physiological models:** Developing parameterized crop models for different crops at various growth stages to adjust VPD thresholds and target soil moisture levels in accordance with specific crop water requirements at different phenological stages.

- Solar energy harvesting: Integrating a small photovoltaic panel and LiPo battery charging circuit to enable fully off-grid operation, making the system viable for deployment in remote agricultural areas without grid electricity access.
- Tensiometer integration: Adding soil water potential (suction) measurement via tensiometers as a complementary soil sensing modality to the capacitive VWC sensor, providing a more direct measure of the energy status of soil water and its availability to plant roots.
- Economic impact analysis: Conducting a detailed techno-economic analysis including cost of implementation, water cost savings, and crop yield impact to quantify the return on investment and build a business case for farmer adoption.

APPENDIX

Appendix A: Bill of Materials with Approximate Costs

Appendix Table A.1: Bill of materials with approximate costs

S. No.	Component	Model	Qty	Unit Price (INR)	Total (INR)
1	Microcontroller	ESP32-S3 Dev Kit	1	680	680
2	Atmospheric Sensor	BME280 Module	1	320	320
3	Light Sensor	BH1750 Module	1	180	180
4	RTC Module	DS3231 Module	1	220	220
5	Soil Moisture Sensor	Capacitive v1.2	2	150	300
6	Flow Sensor	YF-S201	1	280	280
7	Current Sensor	ACS712-5A	1	120	120
8	Motor Driver	L298N Module	1	100	100
9	Mini Submersible Pump	3–6V DC	1	220	220
10	Buck Converter Module	LM2596 5V 3A	1	95	95
11	CR2032 Battery (RTC backup)	–	1	30	30
12	Jumper Wires, PCB, Misc.	–	Lot	250	250
				TOTAL	2,795

The total hardware cost for a single sensor and control node is approximately INR 2,795 (approximately USD 33), demonstrating the highly cost-effective nature of the proposed platform for small-scale agricultural deployments.

Appendix B: Key Firmware Code Snippets

B.1 VPD Computation Function (C++ / Arduino):

```
float computeVPD(float tempC, float relHumidity) {  
    float svp = 0.6108 * exp((17.27 * tempC) / (tempC + 237.3));  
    float avp = svp * (relHumidity / 100.0);  
    float vpd = svp - avp;  
    return max(0.0f, vpd); // VPD cannot be negative  
}
```

B.2 Predictive Depletion and Scheduling Decision (C++ / Arduino):

```
void evaluateIrrigationDecision() {  
    float vpd3h = getRollingMeanVPD(3); // 3-hr mean VPD (kPa)  
    float adi = 0.6*(vpd3h/2.5) + 0.4*(hCum/22.0);  
    float dr = 1.8 * adi + 0.2; // depletion rate (%VWC/hr)  
    float tDeplete = (soilVWC - THETA_CRIT) / dr;  
    float tToWindow = hoursToNextWindow(); // from DS3231 RTC  
    if (tDeplete < tToWindow && soilVWC < THETA_CRITICAL_ALERT) {  
        activateIrrigation(computeDuration(soilVWC, getTargetVWC(vpd3h)));  
    } else if (isInsideIrrigationWindow() && soilVWC < getTargetVWC(vpd3h)) {  
        activateIrrigation(computeDuration(soilVWC, getTargetVWC(vpd3h)));  
    }  
}
```


Appendix C: Pin Configuration Table

Appendix Table C.1: ESP32-S3 GPIO pin configuration

ESP32-S3 GPIO	Connected To	Interface	Direction
GPIO21 (SDA)	BME280, BH1750, DS3231	I2C	Bidirectional
GPIO22 (SCL)	BME280, BH1750, DS3231	I2C	Bidirectional
GPIO34 (ADC1_CH6)	Soil Moisture Sensor (analog)	ADC	Input
GPIO35 (ADC1_CH7)	ACS712 Current Sensor (analog)	ADC	Input
GPIO25 (interrupt- capable)	YF-S201 Flow Sensor (pulse)	Digital / Interrupt	Input
GPIO26	L298N EN_A (motor enable)	PWM / Digital	Output
GPIO27	L298N IN1 (motor direction)	Digital	Output
GPIO14	L298N IN2 (motor direction)	Digital	Output
GPIO2	Status LED (onboard)	Digital	Output
GPIO0	Physical reset / override button	Digital w/ pullup	Input

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